ActInf Livestream #013.0: "Cybernetic Big Five Theory with the Free Energ

Livestream | 54:51

hello everyone
i think we are live this is active
inference live stream
13.0 it is january 3rd
2021 welcome to the active
inference lab everyone new and old
listeners
we are an experiment in online team
communication learning and practice
related to active inference you can find
us at our website
on twitter our email address youtube
page or key base public team and
username
this is a recorded and an archived live
stream so please provide us with
feedback so that we can improve on our
work in the live chat or in comments
and all backgrounds and perspectives are
welcome here
video etiquette for live stream mute if
there's noise in the background
raise your hand so we can hear from
everyone and use respectful speech
behavior
so happy 2021 everyone
for the scheduled active inference live
streams in 2021 we're going to do
every single tuesday from 7 to 9 a.m
pacific
time which will include daylight savings
time changes
and for schedule and participation
instructions just go to this
shortened link here also we're going to

do several special sessions for example
coding sessions or discussions with
external people from other communities
and just if you have an idea for an
event that you think could be
interesting
or you would like to moderate or just
suggest you can let us know or you can
send a tweet and tag that person if you
know them
and then we can set it up but the
regular
active inference live streams the group
plan discussions
are at again that shortened link and it
has
instructions for how to let us know you
want to participate in any of these
events and the first two weeks of
january so these next two weeks
we are going to be in uh reading this
paper
by adam saffron and colin deyoung and
then you can see whatever papers are
most updated for the next several months
at that link today in
active inference live stream 13.0 the
goal is really to set context for 13.1
and 13.2
so kind of like a read-along for the
paper if you want to participate in the
discussion or if you want more context
on it
and the paper is integrating cybernetic
big five theory with a free energy
principle
a new strategy for modeling
personalities as complex systems by

saffron and
young on october 16 2020 at this psi
archive link and the video just like all
the point zeros are
is just an introduction to the ideas and
the context it's not a review or a final
word
the punchline of the paper is that we
can combine psychology with the
cybernetic theories
of cb 5t the cybernetic big 5 theory
which
we'll discuss and free energy principle
active inference to have a better
understanding of psychobehavioral
dynamics in humans and other systems
the sections of 13 will be as follows
first will be
the aims and claims of the paper then
the keywords
abstract and roadmap then go through the
figures and the table
and then conclude with several questions
that people can ask themselves whatever
level of familiarity they have with the
ideas so in 13.1
and 13.2 we're going to be discussing
this paper as a group
so save and submit your questions and
get in touch if you want to participate
okay the paper is as said on the
previous
slide the aims and claims of the paper
in the author's words are
from the conclusion we have begun to
outline how artificial agents based upon
free energy principle and active
inference might be used to test

cybernetic big five theories
uh theoretical propositions regarding
the functional significance of the
dimensions
in the big five trait hierarchy which is
from psychology
here are the implications and why it
matters from the
author's conclusion as well if deep
correspondences can be found between
cb5t
and fep-ai as we suggest is likely to be
the case due to their common cybernetic
foundations
then this integration may allow
personality modelers to obtain new
understanding with respect to functions
and then to parlay that understanding of
function into new hypotheses about
mechanisms
in the brain and finally
identifying the configurations of
personality mechanisms that correspond
to effective cybernetic function
may yield insights into how personality
imbalances can lead to mental disorders
thereby furthering the aims of
computational psychiatry and other
attempts at developing
theories of human mental health and
well-being so
that is what they have set out to do and
by the conclusion that's where they want
to have
gotten and these dot zero videos
are meant to be a sort of bi-directional
on
and off ramp for those in the active

inference community it's
a chance to see the fields that active
inference is adjacent to
and for those who might be learning
about active inference for even the
first time
it could still possibly be a way to
learn about this
broad area from wherever you are
as far as familiarity with these
keywords which include active inference
in the free energy principle
but also generative models cybernetics
and the cybernetic
big five theory and then there's a few
related keywords as well
so first active inference and i
got some interesting feedback and just
thought about it over the new year and i
think it's a great exercise to
always start with just what active
inference is just like how
evolutionary biology which is in the
area i'm in
often talks will begin with the
principles of natural selection
because they're then going to be
demonstrated in the subsequent sections
so
even just having a high level
summarization allows
an individual to personalize their
approach to
portraying what active inferences in the
context of a specific system
or question just like an evolutionary
biologist doesn't need to re-justify all
of natural selection

they just draw upon those principles and
outlines and then apply them
i'm going to start with active inference
in the words of the author whose paper
we're reading
and they wrote work within the free
energy principle paradigm
has yielded a normative model of
behavior called active inference
the active inference framework describes
processes by which free energy is
minimized and provides close ties to
other frameworks such as expected
utility theory and risk sensitive
control
further advances in deep reinforcement
learning appear to be converging on the
kinds of solutions that are predicted to
be necessary for bounded
optimal decision making and learning in
the free energy principle and active
inference framework
and another quotation that they have
about active inference
is the development of the active
inference framework has made connections
between free energy principle
and cybernetics more obvious as it
highlights that organisms are motivated
not only
to predict their perceptions accurately
but also to influence states of the
world in order to ensure that
perceptions correspond to goal
attainment
so that's two really nice quotes on
active inference from the authors
and here's just a way that i kind of

laid it out here maybe we can even try
a new one every single time or something
similar
active inference is a framework which
means that it's a
cluster of scientific theories which are
ideas that make specific testable
predictions hypotheses about the world
that's a theory
and tools so that includes software
toolkits and vocabularies and
shared narratives so it's a framework
it's beyond just a single hypothesis but
there are important hypotheses that are
more or less central to what it
says in specific cases it's a framework
that integrates action
and perception so action is whatever the
behavior of the system of interest is
so if we're talking about a certain
bacterial possibility and affordance
that a bacteria can undertake
then we're talking about action being
our measurement
of something that's for example moving
or being secreted in the world
or emitted from a hidden state and
perception
is sort of the obverse of action in some
way and it relates to
the sensory capacity of the system of
interest
and as we've talked about in other
discussions
there's something like a interface being
defined with the organism and the
environment so that's how we're
integrating internalism and externalism

reductionism and holism all these things
are coming together
because by defining the interface that
one is studying the blanket of some type
you're at the same time delimiting the
system from
the external part so you're not assuming
that there's another level of
boundedness like a closed system beyond
the refrigerator and the kitchen
you can now just model it as your
inference on a certain boundedness given
the measurements and the structure
of the system little digression it
integrates action and perception
downstream of the free energy principle
free energy principle is a statistical
physics approach to modeling
complex multi-scale systems so it uses
several different threads of physics and
for those who are just learning about it
there are
a lot uh better resources than here to
hear specifically about
the mathematical details in this
discussion
but we hope to be able to provide easy
entry points for this
aspect of the discussion as well and
it's
about how far from equilibrium systems
work so things that are
passive like let's just say a cloud of
gas
accreting in some galaxy somewhere
it can appeal to just gravitational
forces to
describe the dynamics through time

however there are systems whose dynamics
through time
like organisms uh their dynamics through
time do not resemble
that of passive bodies and so there's
these active systems
and what is it that these active systems
these mega entropic systems
organization uh in the world how is this
happening
what is life that's where that element
comes in but those are the systems that
we want to understand design predict and
control so active inference is a
framework that integrates action and
perception
downstream of the free energy principle
to explain how far from equilibrium
systems work
what about generative models generative
models can be contrasted with
descriptive models
descriptive models go from data which
are observations about the world
to or results of an experiment and go
from data
to then the modeling of parameters about
the world so
given the height of these 50 children
what is the mean
and the variance of height in the
classroom and so statistics can be
seen as a specific kind of dimensional
reduction about measurements in the
world
for example you have 50 measurements
that's a 50 dimensional vector
and then it's being reduced onto a mean

and a variance component so two numbers
it's like a bottleneck that only two
numbers get to pass through
so it's a dimensional reduction and then
you hope that those dimensions
capture the average value
of the trait or whatever and the
variability
so that's going from data to the model
dimensional reduction
generative models go the other direction
which is
from the model's parameters for example
the mean and the variance
to the expected data given a mean and a
variance estimate for height the two
numbers
can we generate sets of predicted
measurements that are consistent so
generate 50 a draw of 50 children and so
even if it's 51
uh it's going to be just changing it a
little bit and the average is going to
recapitulate those two numbers but it's
going to be
possible to generate hundreds or
thousands or counter factual worlds
given this generative model
generative models are useful for
statisticians
for a lot of the same but also some of
the different reasons that they're
different
or that they're useful for organisms to
have generative models
so here's some of the similarities is
that in both the statistician and the
organismal case

it allows the system to use a lot of its
previous knowledge
on sparse data sets because generative
models
might only need a slight cue to tip
their likelihood from something very
unlikely to totally likely for example
hearing uh the sound of an ambulance
outside your window means that it's
extremely likely that there is an
ambulance there
even though it's also possible that
there's one at other times it makes it
extremely likely
and one kind of model that's used that's
a generative model
and a way just to understand without the
control theory without the cybernetics
how a generative model is useful
statistically is what's called
expectation maximization and this cycle
expectation maximization the data or
initial estimates are input
into what can be thought of as one layer
of the model which is just the estimator
of the observed data parameters
and parameter is just in the
computational sense any
single variable that's going to be
stored so these are the the data this is
the actual 50
children in their heights and then
there's a maximization step
which asks what is the maximum
likelihood of those underlying hidden
parameters
which can be a different dimension than
this one and then

there's a generation step which is the
expectation step which data are expected
uh or expectable i guess i don't think
that's a word but that's kind of it
given the hidden parameters that are in
the world
and so this can continue until it
converges
and um this could be about
static things in the world or it could
be about dynamic things in the world
and the expectation maximization could
also be on
um causal things in the world and when
that inference is going to be about
action that's where action enters into
the loop because instead of just having
a data set and trying to converge on the
best descriptor
it's uh trying to converge on a policy
that could involve
hidden aspects of the world so let's go
a little bit more into detail in this
generative models and processes
generative models and generative
processes and active inference idea
because
i don't think this uh person's blog has
been
on here chuck tutorial on active
inference some really nice images
so here on the left is this idea that a
generative process which has a true
hidden state
s is emitting observations which
are modeled by some p function but its
observations are being emitted
from the world and the niche is really

the generative process that leads to
those observations
like the sky emits the sun and the moon
or the ecology
emits this rainfall or something then
on the other side of the blanket we have
the organism
which is embodying and enacting a
generative model
of those observations so that's like
your brain
predicting what the retina would see
well how do we know that the brain is
predicting what we would see there's no
blind spot
it has usual under healthy conditions
equal clarity everywhere and equal color
vision everywhere
and that's just not uh concordant with
the actual distribution of cells on the
retina
so really it's like upgraded resolution
at your periphery
that's an illusion based upon the
generation of the assumption
that things are actually equally clear
outside of your perception so it doesn't
uh have the same actual experience
uh characteristics as what you'd expect
in the retina
just you know interesting things and you
can imagine that for other senses too
so the organisms enacting this
generative model of those observations
and then the surprise or the residual of
those predictions is kind of getting
passed upwards into these models
and that's changing how the model of the

organism
in this bayesian framework here is
uh modeling these observations so the
organism has a generative model or is a
generative model depending on how you
think about it
of the niche which is what facilitates
action if your model of the circadian
rhythm
is 18 hours and it's 24 hours for the
real thing to happen
then your niche your temporal modeling
is off
and it's going to be pathological and
maladaptive or potentially not
that's where selection comes in
so here's another uh representation from
that same blog that
shows how now from that inference
process described here where the
organism has a generative model of
inference on those observations
in the world so i'm predicting what
observations on bitcoin price i'm going
to see
emitted by the true generative model now
i don't know what the generative model
is but i can
see the observations the organism
has preferences for certain observations
so here the darker
circle is representing preferable
observations
so the observations in this case are
enteroceptive which is really
nice and um the organism has a
preference for let's just say
in this categorical example one outcome

of the
observations over the other the gray one
and now
we can make a two by two matrix with the
two possible observations of the world
with a preference over them and the
state of the world which can't be
directly perceived which is whether the
stomach is empty or full
this isn't a physiological uh debate
it's just that's the simple
case here and that relies on this
a matrix which is the probability of
observations of
the world which is the data
given the states which can't be directly
experienced which is the stomach
that's the a matrix then the b matrix
maps a policy in the prior states
to the way in which the hidden states
are expected to change through time and
you can evaluate it for different b's
so here there's two possible u's which
are policies
eat or do nothing so
under under each policy eating or doing
nothing
which are represented by two different
matrices b1 and b2
the state at time t is like the current
moment
and in the state at time t plus one is
this day at the next moment
so there can be mappings between the
present and the future
conditioned on policy that's the
probability
basically of the state uh right now

given the state in the past and a policy
so that's phrased from like past to now
and this is about now to the next step
and here is the graphical representation
through time so t minus one is the past
t is the present moment and t plus one
is the future so the actual state of the
world is evolving under a certain
dynamic and it's emitting observations
and then under multiple possible policy
trajectories policies uh parameters or
different policies that can be
enacted and embodied what is being
estimated
is this relationship from how previous
states
of the policy influenced or were
associated with this true state
of the world at the current moment in
this layout
and how that observation then influences
uh the updating of the policies in
uh relative comparison for the current
moment moving forward
under a generative model of the action
of action in the niche by the organism
all right big five psychology so the big
five are dimensions of variation
that are statistical and usually from
self-reported surveys that are based
upon basically personality differences
in the population
so it's a dimension of variability in
populations
kind of like the trait of heritability
from quantitative genetics
so it's a measure of variance explained
on a certain population sample

it's not a causal model of traits nor is
it a fraction due to genetic
causes even though that fallacy is made
throughout the literature
so the big five is based upon the use of
statistical methods
basically correlation matrices and
principle component analyses
on survey data that finds patterns
across questions and participants
and the acronym for big five is ocean
so here's an example of how the big five
are calculated and then what the oceans
stand for
so what will happen will be that a bunch
of participants are
asked a survey or are scored through
some
way and this survey may have a hundred
questions or a certain
different number of questions and that
corresponds to
a hundred dimensions of data it's an 100
dimensional vector per person
so the total survey study design can be
um just the columns are the people and
the rows are
the questions and how they answered and
so that's a simple
matrix that can be given all the data of
that entire survey
and some statistical techniques are used
on that
matrix and like a principal component
analysis
and it according to the psychology
literature
falls out into these five large axes of

variation so when we do the pca
the algorithm is not told specifically
which questions are expected to be
correlated with
or negatively or positively with other
questions
but because there are actual statistical
regularities in certain
pairs of questions or sets of questions
in the real world in terms of how people
answer things for example a question and
its negation should have a negative
correlation
um as well as other relationships in the
world uh it's possible to find the
structure of some of these
uh dimensions that map very well on
these dimensions of variability
in the data and so the five big axes of
variation
are openness to experience the axis
between inventive and curious versus
consistent consciousness
conscientiousness extroversion
agreeableness in neuroticism or ocean
and uh i uh hesitate to go too much into
the adjective
side because those are always very
english language dependent
and really it's about the structure of
the data in the real
observations and here's just a few other
issues with
uh the topic of big five which is that
there's often population and survey
specific covariance structures
though some structures are quite well
correlated across different surveys

there's often a normative or a
culturally narrow
understanding of what personality is or
how it manifests and how it represents
itself
and also there's just this fundamental
difficulty in attaching the natural
language
words to these differences in the
manifold
um one way though that we can kind of
get out what these dimensions really are
is by looking at the questions that are
very highly correlated with each of
these axes
so for openness a few questions i drew
three questions from each of the
letters is i'm quick to understand
things
i like to try new things i have a good
imagination
c conscientiousness is i'm always
prepared i pay attention to details i
like order
extra version e is i am the life of the
party i feel comfortable around people
i start conversations a agreeableness is
i take time out for others i feel others
emotions
i make people feel at ease neuroticism n
is i get upset easily i have frequent
mood swings
and i worry about things such as in my
life
and one interesting note on this paper
and definitely
a little fun fact
is that um one can do a vector

decomposition on the big five to reveal
these additional aspects of personality
beyond those five
principle structurings that's of course
graining to a very
high degree and this is going to go one
layer chiseling down into the
statistical variability structure
and this paper i was searching in my
library for it because i
thought it was an interesting one and i
remembered it and it turned out that
it was actually the author of this paper
from 13
14 years ago with jordan peterson yes
that jordan peterson and it draws
somewhat on this idea of vector
decomposition but statistical vector
decomposition
and just as refresher uh two vectors can
be added
kind of head to tail and they end up at
the same
place no matter which way you add them
just like
regular numbers like two plus three
equals five and three plus two equals
five
so they took the five
big dimensions and broke them down the
the blue and the red arrows they they
don't
up and the down they don't represent
anything about the trait characteristic
it's just using the same example of
vector addition it's breaking down
a statistical dimension into another
axis of variability

which is a pretty cool technique and
it's going to relate in the paper that
we're going to discuss
to the ways to get at very fine-grained
aspects of the high-dimensional space
that represents action
propensities cybernetics is uh it's not
the first time we've had cybernetics as
a keyword on the active stream
but each time we do want to kind of see
it a little bit differently
so the image on the left is a brain with
a bunch of wires and kind of represents
this crossover
mind machine mapping is it an analogy is
it an interface
is it a vision is it is it a prophecy
all these different things that mind and
machine can be
is kind of captured in cybernetics
culturally but also on the right side
is a more formal systems diagram
approach to cybernetics which describes
kind of first and second order
cybernetic
systems and here is from this cybersoft
post down on the bottom they wrote
the basic idea of cybernetics is that
complex systems such as living organisms
societies and brains are self-regulated
by the feedback of information
by systematically analyzing the feedback
mechanisms which regulate complex
systems
cybernetics hopes to discover the means
of controlling these systems
technologically
and to develop the capability of

synthesizing artificial systems
with similar capacities there's a few
other parts about cybernetics that are
interesting
because it has had quite its own history
and it's uh related to complex system
study but there's some interesting
history so here's a few
more quotes from this article they write
the word cybernetics was coined by the
mit
mathematician norbert weiner in the
summer of 1947 to refer to the new
science of command and control in
animals machines
which he helped to establish and develop
the word was developed derived from the
greek kubernetes
meaning steersman or ship pilot unknown
to weiner at the time
plato had used the adjective kubernetes
in the gorgius to refer to the science
of piloting
and the french physicist marie ampari
had derived the french word cybernetic
direct directly from the greek to refer
to the science of government
in his classification of sciences and so
the word kuber leads us straight to
governor
the regulator guide and pilot of steam
engines and states
it thus lies at a powerful metaphorical
intersection of political economy and
technology
social and material order and the
regulation of animals and machines
and then another quotation that was

pretty interesting from the end of this
article was when donna haraway
challenged feminists and leftists to
embrace the language and metaphysics of
cybernetics what was she aiming at
was the cyborg just another metaphor for
hybridity
if it was then why was she so insistent
on adopting not only the terminology of
cybernetics systems and information but
its whole ontology
no this was not mere rhetoric this was
this move was political and it was savvy
wise and insightful
for haraway shows us it is not just the
world the environment that is cybernetic
we are cybernetic we are cyberworks
cybernetic organisms
we are hybrids made up of organic beings
and mechanical parts
and the only thing that ties us all
together you and me you and other parts
of you is information
fundamentally cybernetics is a new
ontology andy pickering has called it
an ontology of becoming the universe is
not made up of matter
particles or energy nor is it made up of
ideas words and symbols
it is not numerals or universals or
plenum or atoms or modads
the world is made up to feedback loops
of information
so pretty cool quote
cybernetic big five theory cb5t is a
line of research pursued by one of the
co-authors of this
paper primarily colin deung and it uses

cybernetics
as the name might suggest to integrate
the dynamics of personality and get at
potential mechanisms amongst other
things
and the highlights from the 2015 paper
which is cited in the 2020 paper
they write cybernetic big five theory is
a new integrative theory
of personality personality traits
reflect parameters of involved
cybernetic mechanisms
characteristic adaptations which is a
term from cybernetics our goals
interpretations and strategies in
updateable memory specific mechanisms
are identified for traits in the big
five hierarchy
implications for pathophysiology
psychophysiology
and well-being are discussed and this
was a figure
from the young 2015 paper but it's
actually drawn
from jordan peterson's 1999 book maps of
meaning
and here's how it is in the book slash
paper
where at any given state there's the
current state of the world
and strategies are being implemented to
attain desired states or goals then
anomalous information is always entering
the picture from chaos entropy the
unknown
at some point the current state of the
world given the strategies and the
desired goals

can no longer keep the integration
process
maintained the center cannot hoddle and
there's a disintegration
process that leads to a state of chaos
in the entropy in the unknown
followed by a reintegration because you
can only
have a reintegration from disorder to
order you can only reintegrate even if
it's a very similar state
because it's still an accessible one
even if it's a similar one it still is a
reintegration
and if the world state has changed that
means the niche is different which means
the reintegration must be different
so i thought it was kind of interesting
to map this onto a few other ideas from
active and other things we've
sorry that we've talked about in the
streams
so first there's the dividing line
between order and disorder which
correlates to basically the blanket the
interface the whole thing we're talking
about with the self-organizational
systems
because that boundary between the agent
and the niche
again is not just an informational one
it's also an info thermodynamic one
so the agent is where organization is
happening for example
the uh person in their apartment is
eating food
and they're organizing biological matter
radiating heat they could be getting

more organized making more organized
tissue of any kind depending on their
hormonal state
and then depending on that the room is
going to be warming up and there's going
to be
heat released in the supply chain so
that's actually like a thermodynamic
boundary
as well as an information one and we can
think about these
preferences over different observed
outcomes of the world
as being akin to goals peterson's
framework here is still
very reward oriented rather than
precision oriented that does enter the
picture here we'll get to that in a
second
the strategy is like policies which
truly are strategies
and the current state of the world and
the interpretation are like hidden state
estimates
which could be like is my career stable
or will things be okay with this
thing of the world so the current given
current
understandings of the world which can be
accurate or not helpful or not two by
two
and given strategies being implemented
which is like the field of affordances
and active inference
are organisms attaining their preferred
states their desired states of the world
and in the framework of peterson as well
as an active inference anomalous

information is continually
not just occasionally entering into the
picture so it's always a question of
resiliency and of self-organization
around anomalous information being
integrated because
no uh model is going to be perfectly
predicting it's about
incorporating the variability into a
healthy and
resilient model and here it's like
surprise being passed from the
environment
through the interface to the organism
and the organism can disintegrate and
enter into
potentially a pathological region or a
strange attractor state that could
re-integrate or re-stabilize in a newer
different niche
into another different state policy
preference
combination and this
encounter with chaos which could be the
dark knight of the soul or the hero's
journey
or alchemy or phase transitions of the
self
that's an interesting thing to think
about with the self organization and the
reorganization reintegration of
societies
cities teams relationships individuals
another thing that is mentioned in the
paper but it wasn't keyword
was spm which is statistical parametric
mapping spm so here's from the spm
website whose url is on the bottom

it says spm12 first released first of
october 2014
but previous versions of spm have been
released for many years and last updated
on the 13th of january 2020
so almost one year old at this point is
a major update to the spm software
containing substantial theoretical
algorithmic structural and interface
enhancements over previous versions
cool what is spm spm
is a software package that has been
designed for the analysis
of basically brain imaging data
sequences but it turns out that that's a
quite general
question for a lot of reasons which
we'll discuss a few of right now
so what spm does is it starts with a
series of
images that are from various types of
experimental designs
it could be from similar or different
cohorts or it could be
different kinds of time series from one
subject
and the current release is designed for
the integrated analysis of fmri
pet spect eeg and meg so the most
common interfaces for neuro imaging
are incorporated uh despite their
usually extremely different error
profile
and actually uh representing quite
different aspects of the tissue
being imaged and over different time
scales with different accuracies so
it's quite a general question to

integrate different kinds of time series
into underlying generative models of
extremely specific
types what is the statistical parametric
map it is the construction and the
assessment
of a spatially extended statistical
process so a
statistical process that happens through
time and space
that is used to test hypotheses about
functional imaging data
so there's like the map and the
territory that's the mapping is kind of
that two layer the data to the
underlying model that's the mapping
and then it's a statistical parametric
mapping
so like a p-value comes from a z-score
or t-score
or some other type of variable that's a
statistical parameter
and so it's going to be like an
underlying statistical test and
it's related to a lot of different kinds
of data
and this spm textbook from 2007
is uh quite a mathematical read
maybe it or a later wikipedia
version of the spm toolkit could be
interesting to have a math
set of discussions but we have this one
at home and it's a classic
it's a true true classic here's just
a summary of what spm does so that
the key aspects can be seen in this is
what fristen
for years before becoming more popular

in the machine learning and uh theory of
everything
and philosophy of everything universe
uh many years before that he was and
still is
the most cited neuroscientist and the
huge
reason for that is the many many papers
that have gone into this type of
development and then also its use as a
toolkit
broadly by many communities so the way
that it begins with
the spm is taking in the data which
represents
slices of different underlying objects
so it can be attained measured in
various different ways whether it's all
the measurements simultaneously
or whether the slices take time as they
slice through a system
and that is first done uh there's an
inter
first processing step that's basically a
realignment to
a standardized brain atlas or a
person-specific brain atlas for example
so you can take a static image that has
much higher resolution and align
dynamical images
or not to this more static high
resolution image which can help a huge
amount
that is then processed through smoothing
which is like this thing called a kernel
which is like you have a fabric which is
the data and there's going to be like a
finger and it's going to move across the

data and re-weight the data
according to some smoothing technique
and
this represents a gaussian kernel so
this is just
blurring the image to ease out a little
bit over the noise at a certain
spatial and temporal scale that can also
be related to
specific instrumentation related biases
that are known
like non-linearities in the response and
that
is this process um where the the data
are
processed into um a cleanly comparable
format spatially and temporally and
statistically renormalized
then there is the specification of a
design matrix which there's so many of
these in the spm textbook it's really
awesome and they look really beautiful
and the design matrices are uh
very broadly types of things that you
want to consider in your analysis
whether it's something you know about
the patient like their age
or their pathological diagnosis whether
it's something you know about them
and you want to include it in the model
or not uh
as in as something you're predicting or
including in your prediction of
something else
or whether the design matrix includes
other more esoteric components that are
often discussed in this spm textbook
like

these time kernels that design matrix
and the smooth brain image sequences are
put together into a general linear model
and it's funny because you hear a lot of
disdain for linear models but this is
linear models
in a extremely rigorous and non-linear
way
that can lead to these parameter
estimates
because it's like a linear model on a
space
that's been warped by this
prior and updates in a multi-scale way
so that's really the interesting part is
it takes renormalization to the next
level
and then applies general linear models
in an extremely specific way
though i don't understand all the
details that relates to
what is called a statistical parametric
map
which is here like a heat map of um
the uh the score of that
voxel so that's like the z score so like
a z
score of two or two standard deviations
above the mean
or one or whatever it happens to be it's
like a z-score positive and negative
and then there's a variety of approaches
that can be used
to then on this statistical parametric
map
draw p-value
distinctions on this and it turns out
that that

process of setting a p-value threshold
and saying whether 80 of the brain is
used in a task or one percent or
correct for one person's structure but
you're trying to detect the structure
that area is an extremely nuanced field
no pun intended and it draws on this
random field theory and there's several
chapters in the spm textbook that do
random field theory
and it was just like quite an
interesting approach to statistics that
is uh really being implemented here all
right the abstract of the paper is that
cybernetics is a study of goal-directed
systems that self-regulate via feedback
a category that includes human beings
cybernetic big five theory attempts to
explain personality in cybernetic terms
conceptualizing personality traits as
manifestations
a variation in parameters of the neural
mechanisms that evolve to facilitate
cybernetic control
the free energy principle in active
inference framework fep ai
is an overarching approach for
understanding how it is that complex
systems manage to persist in a world
governed by the second law of
thermodynamics
the inevitable tendency towards entropy
so the authors have framed
the conversation as being firstly about
cybernetics
first word that's about action selection
in complex systems that self-regulate
and use feedback

big field then within the context of the
cybernetic big five theory which this is
a psychological theory that draws
heavily on cybernetics
we are going to consider the neural
mechanisms that facilitate
and have evolved to facilitate this kind
of cybernetic control
then the free energy principle and
active inference framework are
introduced
um to explain a similar set of topics
behavior as well as this resistance of
dissolution
far from equilibrium systems although
these two cybernetic theories
were developed independently they
overlap in their theoretical foundations
and implications
and are complementary in their
approaches to understanding persons
fep ai contributes a potentially
valuable formal modeling framework for
cb5t
while cb5t provides details about the
science and structure of personality
in this chapter we explore how cb5t and
fepai
may begin to be integrated into a
unified approach to modeling
persons we'll need another acronym won't
we
so the roadmap the first section
is a cybernetic approach to personality
and has the first figure
which shows some of the causal processes
that are related to the functioning of
personality of the individual

the second section is the free energy
principle and cb5t
third is predictive processing and
personality neuroscience
active inference generative models and
personality modeling
there's two figures in that section
which draw upon
the graphical models some of which we've
seen some of the structure before
and also the spm demo then in section
six it talks about
personality traits from the perspectives
of cb5t
and fep ai and there's a representation
of the personality trait hierarchy
and then the final three sections are
the metatrades
the big five personality traits
then a conclusion on the levels of
analysis in personality
modeling table one then has some of the
analogies
between uh or mappings between cb 5t and
fep ai so figure one
in the paper is a uh drawn from another
paper and it relates to the causal
processes
that are involved in the functioning of
personality within an individual
and so it's a pretty uh loose
chart saying that both genes and
environment
play into these relatively stable
parameters of cybernetic mechanisms
so that's saying somebody who has uh
these are uh cybernetic underlying
features like given a certain attractor

state some people may
spiral towards it or spiral away from it
or stay in a certain stable orbit to it
and that is the cybernetics of
biological systems that maps onto
personality trait architectures
statistically in matrices
because there's this underlying
statistical regularity in the world
and that leads to a feedback system with
stigma g perhaps
of characteristic adaptations and life
outcomes and that's why there's
correlations between
life outcomes and personality traits and
it's hard to determine cause and effect
and
all these other aspects of it
figure two is pretty nice it's the
authors redrawing i believe
of the graphical model structure that
we've seen a lot of
but i wanted to read a quotation from
page 21 because they
laid it out very nicely they wrote
generative modeling in fep ai can be
complex but the basic setup
involves specifying agents and their
situations as systems of interacting
variables and parameters
these variables take the form of
mutually referencing matrices whose
entries
specify probabilistic mappings between
different portions of the generative
model
nice as we previously described a single
objective function of

expected free energy is used that's g to
determine which policies are selected
according to whichever actions are in to
determine which policies are selected
that's pi
uh according to whichever actions are
anticipated to minimize overall
prediction error
an example fep ai generative model may
include the following matrices where
each mapping is conditional upon the
probabilistic mappings from
preceding matrices a a matrix is the
likelihood
mapping from latent world states to
sensory observations or how sensations
are interpreted via perception and
imagination
from the agent's perspective this would
correspond to answering the question
what is likely to have caused my
sensations the b matrix is the
transition probabilities
or how dynamics are expected to evolve
from one moment to the next from the
agent's perspective this would
correspond to answering the question
how do i believe things are likely to
change through time three c
matrix the c matrix is the prior beliefs
over policies or outcomes or which
states an agent predicts itself
occupying
via mechanisms of prediction error
minimization these prior expectations
constitute the attracting states of the
system and function as preferences
from the agent's perspective this would

correspond to answering the question
where or what do i want to happen and
thereby doing
slash enacting it through active
inference in the ways described above
from a cb5t these preferences are an
agent's high level goals
four the d matrix the d matrix is an
agent's belief regarding its initial
position within the decision
environment from the agent's perspective
this would correspond to answering the
question
where am i likely to be in this kind of
situation or where am i likely to have
started
the n5 the e matrix the e matrix is the
baseline prior over policies governing
policy selection in a habitual or
heuristic fashion from the agent's
perspective
this would correspond to answering the
question what am i likely to be doing
right now
and thereby doing slash and acting
through active inference but with
actions deployed in a fashion that
ignores uncertainty
with respect to the likelihood of goal
attainment in a particular situation
from a cb5t perspective these habit
priors would constitute a major class of
characteristic adaptations
pretty nice then you have the states and
observations and then
the beta and the gamma which is the
precision uh
estimators which is the confidence in

the temperature parameter
over that confidence so very nicely laid
out by the authors
and always uh cool to hear it a
different way and framed as a catechism
which is actually like
what does each matrix represent the
question uh
what is the question that reflects how
each matrix is answering something in
the model
doing something in the model figure 3
is an illustrative figure accompanying a
generative modeling demo that is
pre-packaged with spm-12
this particular simulation involved
modeling curiosity with respect to a
task with unknown rules
where agents attempted to maximize
valued outcomes while also needing to
discover
the latent task structure and the
generative
patterns of the data that are part of
this integrated generative model in
spm include the inferred rules of the
world
the top left the patterns of the task
choices here
and the neuronal activity anticipated to
occur
based upon mechanistic process theories
associated with fep ai
bottom left in all these three so unit
firing firing rates
local field potentials and dopamine
responses as
a function of these things happening

under this
rule and choice so it's policy behavior
and neural or underlying uh responses
which can be of
various categories from the neuro
metabolic like dopamine
to the firing the local field potentials
so
it's quite a broad modeling framework
figure four
is the personality trait hierarchy and
they write
note that the structure indicated here
is the between person covariance
structure of traits in the population
not within person causal structure so
in other words these connections reflect
ways that you can partition
axes of variability not like there's
some sort of downward
cause so that's sort of the challenge
with this naming
of these things is that they're all kind
of like adjectives they're all they all
seem like they're equal adjectives
um i mean some might have a more
negative or positive connotation or
preference for some people but they're
all just adjectives
and yet really it's a specific
statistical structure
where the connections don't really
reflect
relationships they're sub-components
that are
distinct under another sub component of
a population variability
so it's really better just to think

about what the real question
is and what this statistical structure
is doing
because the adjective part i just don't
really know how
good of an approach that is and then
also the piece of the niche
is that eco evo devo so ecology
evolution and development
are what structure those stability and
plasticity metatrades
because the ants that we see today for
example
they've been on a lineage that's been
successful since the beginning of ants
and back even further so it has had to
maintain
some parameter values which have kept
pace with
changing local context so that's what
structures
these really specific top level traits
because it's actually the process that
structures the entire landscape that all
these things are playing out upon
table 1 shows some mappings between the
key concepts in cb5t and their
definitions
and the free energy principle or active
inference
terms or related topics so cybernetics
itself
is about governing systems that
self-regulate via feedback
and that's related to many of the
cybernetic underpinnings and also
information theory and entropic aspects
of

fep psychological goals
are representations in cybernetics
that get enacted through action and fep
they would say that that is a bayesian
belief about the expectations
of uh how prediction errors are going to
be
minimized so that's preferences via
policy selection
psychopathology is a persistent or maybe
a
self-diagnosed or not failure to move
towards one's psychological goals due to
failure to generate effective new goals
over different time scales let's say and
that could map
in the fep to a persistent inability to
minimize prediction error
and there's a lot of interesting work on
that in the second part of table one
personality are the reasonably enduring
psychological individual
differences although also it would have
to do with the ecological niche because
it can't be defined abstract from the
relationships and the
experiments and then the fep active
inference correspondence would be these
enduring attractors
that define systems as weakly ergodic
processes so states that some systems
revisit but others don't
so some people return to a youtube
channel or a tv channel or whatever
and some people don't some people it
never enters into their orbit
and so that's why it's always
conditional on what is being received

from the niche
those personality traits are those
probabilistic descriptions of stable
patterns
within personality and their
correspondence suggested for the fep is
the minimal message length or the
maximally informative
descriptions of agents as generative
models so this is like the parsimonious
description
of the agents representing their niche
the characteristic adaptations are the
stable goals interpretations and
strategies specified in relation to
basically the niche and that relates to
this whole policy and adaptive component
of fepai
so that's basically it for this um
video and a few questions to return to
at the end questions that are often
asked what would we really
get at with a good framework here so
what would a good framework enable
what are the unique predictions and
experiments of this model
so what is our knowledge gain what are
we getting out of
this specific integration paper um
and how is it going to influence not
just how we think about the world
but how we act in the world in terms of
which experiments are now optimally
informative
even if those experiments the data
already exist maybe it's just an
analysis
but that's still an experiment another

question is what's the next step for
active inference
maybe we could bring active inference
models
to bear um using some of these new
tutorials that are becoming available
on large psychological databases and see
if there's an active inference or a
cybernetic descriptor of behavior
that explains more variance more
parsimoniously than the big five
structure does another question is what
is the goals
of this research what do different
people want to see
happen and what are the goals of the
research
and also very importantly what are we
still curious about
because there's going to be 13.1 and
13.2
where we as a group discuss this paper
and then we'll have more papers to come
so it's okay to be curious about it
any level anytime you're listening to
this
live or in replay but yeah just uh
leave a comment on the video or join us
in the discussion if you listen to this
at the right time
and it's good to hear about what you're
curious about because in this video i
really just tried to
run through the figures and give some
background or some definitions so maybe
some parts were
a bit too much or too little for some
people so as always it's just throwing

it out there
and let me know uh how can improve
so thanks for participating even
listening is participating
we provide follow-up forms to live
participants
and we request your feedback suggestions
or questions
and just uh stay in communication and
hopefully
we will see you another time so thanks
for listening and i'm gonna stop the
live stream